

Course 3: Project Planning Putting It All Together

Module 1: Beginning the planning phase

- Schedule: The project timeline, which includes the start date, the end date, and dates for events in between
- Budget: The budget accounts for the total cost to complete the project
- Risk management plan: Searching for possible problems related to the project and planning ahead to mitigate these risks

Project kick-off meeting

The first meeting in which a project team comes together to ground everyone in a shared vision, gain a shared understanding of the project's goals and scope, and to understand each person's individual roles within the team.

Who to invite to a kick-off meeting

- Project team
- Project stakeholders
- Project sponsor

Kick-off meeting goals

- Establish a shared vision
- Align on scope
- Build team rapport
- Ask questions and offer insights
- Set expectations

Kick-off meeting agenda

- Introductions
 - Team member names
 - Project roles
 - Fun facts
- Background
 - How the project came to be
 - Why the project matters
 - Set a shared vision
- Goals and scope
 - In-scope
 - Out-of-scope
 - Target launch date
 - Milestones

- Roles
 - What work everyone is responsible for throughout the duration of the project
- Collaboration
 - Shared project tools and documents
 - Communication expectations
- What comes next
 - Set expectations and action items
- Questions
 - Gain clarity on meeting topics
 - Ensure the project benefits from diversity of thoughts experiences, and ideas

Kick-off meeting best practices

- Ask a teammate to take notes on key points and action items
- After the meeting, send a follow up email that summarizes the key points and outcomes from the meeting, and any action items to the attendees
- Invite attendees to reach out if they have any additional questions
- **Set the right time.** Choose a meeting time that works for everyone. Be mindful of time zone differences.
- **Set the right length.** Choose an appropriate meeting length—no more than one hour. You don't want to waste people's time, but you also don't want to run out of time. Kick-off meetings work best when you first share key information and then spend any additional time on questions and team building.
- **Invite the right people.** Be strategic about including the appropriate people. The goal is to invite attendees who play a role in the development and execution of the project, such as all team members, stakeholders, and the project sponsor. You don't want to leave anyone out, but you also don't want to invite people who shouldn't be there.
- **Designate a notetaker.** The discussion that takes place during the meeting is important. It is critical that you document any feedback, changes, or questions asked by attendees. If you are leading the meeting, designate someone else to take notes before the meeting starts. You can also use tools like Chorus Notetaker, Google Keep, Google Docs, or Microsoft OneNote.
- **Set the agenda.** To recap what we discussed in the video, a kick-off meeting agenda should generally include: introductions, the project background and purpose, project goals and scope, roles and responsibilities, the collaboration process and project tools, what comes next (expectations and action items), and time for questions and discussion.
- **Share the agenda.** Prior to the meeting, share the agenda with attendees via email and identify speakers for each topic. By sending the agenda in advance, everyone will have an idea of what to expect, time to prepare for anything they may need to present or discuss, and time to generate questions or ideas.
- **Stick to the agenda.** During meetings, discussions can sometimes go off topic or take longer than expected. As a project manager, it is your job to keep the meeting on track by redirecting discussions to the items on the agenda.
- **Follow up after the meeting.** After the meeting, make sure to send out a meeting summary featuring the meeting notes and any action items.

Project Milestones and Tasks

- **Project task:** An activity that needs to be accomplished within a set period of time and is assigned to one or more individuals for completion. The work of a project is broken down into many different project tasks.
- **Project milestone:** An important point within the project schedule that usually signifies the completion of a major deliverable. Milestones are significant checkpoints in your project, and keeping track of them helps ensure that your project is on schedule to meet its goals.

Setting Milestones

- The first step to setting a milestone is to evaluate your project as a whole.
- To get a good sense of timing, you can connect with teammates to discuss the tasks required to reach each milestone, and get their estimates for how long those tasks will take.
- Once you've determined your milestones, the next step is to assign each one a deadline.
- When determining deadlines for milestones, you'll also want to consider the needs of your stakeholders.

Things to avoid when setting milestones

- **Don't set too many milestones.** When there are too many milestones, their importance is downplayed. And, if milestones are too small or too specific, you may end up with too many, making the project look much bigger than it really is to your team and stakeholders.
- **Don't mistake tasks for milestones.** Remember that milestones should represent moments in time, and in order to map out how you will get to those moments, you need to assign smaller tasks to each milestone.
- **Don't list your milestones and tasks separately.** Make sure that tasks and milestones can be visualized together in one place, such as a project plan. This will help ensure that you are hitting your deadlines and milestones.

Scheduling Approaches

- **Top-down scheduling:** In this approach, the project manager lays out the higher-level milestones, then works to break down the effort into project tasks. The project manager works with their team to ensure that all tasks are captured.
- **Bottom-up scheduling:** In this approach, the project manager looks at all of the individual tasks that need to be completed and then rolls those tasks into manageable chunks that lead to a milestone.

Work Breakdown Structure (WBS)

A tool that sorts the milestones and tasks of a project in a hierarchy, in the order they need to be completed.

After completing a work breakdown structure, you should have:

- Sets of discrete project tasks that ladder up to each of your milestones
- Team members assigned to each task

Assigning Tasks

- Tasks are typically assigned according to a person's role in the project.
- To assign tasks between two or more team members with the same roles, take into consideration each person's familiarity with the tasks at hand.

- Consider each teammate's workload.
- Ensure that teammates are clear on their assigned tasks.
- Assigning tasks creates a sense of personal responsibility for members of the team.

Module 2: Building a project plan

Components of a Project Plan

- Tasks
- Milestones
- People
- Documentation
- Time

Time and Effort Estimation

- **Time estimation:** A prediction of the total amount of time required to complete a task.
- **Effort estimation:** A prediction of the amount and difficulty of active work required to complete a task.
- **Sub-tasks:** Smaller tasks that are required to complete a larger task.
- **Unrealistic effort estimates** happen when you've underestimated the amount of time it'll take to complete a task.
- Too much optimism can lead you to overlook potential risks that could set your plans behind schedule.
- Your teammates will have the most realistic understanding of the amount of work required to complete a task, and will be able to provide you with the best estimate.
- Asking the teammate assigned to the task for their estimate is likely to yield a more accurate estimation.

Scheduling Buffers

Buffer: Extra time added to the end of a task or project to account for unexpected slowdowns or delays in work progress.

- **Task buffers:** Extra time tacked on to a specific task.
 - Should be used primarily for tasks that are out of the project team's control.
 - Should be used more sparingly for tasks within the project team's control.
 - Adding a buffer to every project task could lengthen your project schedule unnecessarily.
- **Project buffers:** Extra time tacked onto the end of a project.

Capacity Planning

- **Capacity:** The amount of work that the people or resources assigned to the project can reasonably complete in a set period of time.
- **Capacity planning:** Refers to the act of allocating people and resources to project tasks, and determining whether or not you have the necessary resources required to complete the work on time.

Critical Path and Float

- **Critical path:** The list of project milestones you must reach in order to meet the project goal on schedule, as well as the mandatory tasks that contribute to the completion of each milestone.
- Your critical path includes the bare minimum number of tasks and milestones you need to reach your project goal.

- To identify the critical path and manage schedule:
 - Identify which tasks can happen in parallel vs. which tasks can happen sequentially.
 - Determine which project tasks have a fixed start date.
 - Determine which project tasks have an earliest start date.
 - Identify if a task has float (also called slack).
- **Float:** Refers to the amount of time you can wait to begin a task before it impacts the project schedule and threatens the project outcome.
- Tasks on the critical path should have zero float.

Soft Skills for Project Planning

- **Soft skills:** Personal characteristics that help people work effectively with others.
- **Soft skills for accurate estimation:**
 - Asking the right questions
 - Negotiating effectively
 - Practicing empathy
- **Open-ended questions:** Questions that cannot be answered with a "Yes" or "No". Examples:
 - How long does it typically take you to mock-up a website design like this one?
 - How complex is this task?
 - What are the risks associated with this task?
 - When do you think you could have this ready?
- **Empathy:** A person's ability to relate to the thoughts and feelings of others.

Project Plan Best Practices

- An anchor of a good project plan is a clear schedule, containing all the tasks of a project, their owners, and when they need to be completed.
- Understand the expectations, priorities, risk assessments, and communication styles of your stakeholders and vendors.
- **Best Practices:**
 - Carefully review deliverables, milestones, and tasks
 - Give yourself time to plan
 - Recognize and plan for the inevitable: things will go wrong
 - Stay curious
 - Champion your plan

Project Schedules and Work Management Tools

- **Gantt chart:** A horizontal bar chart that maps out a project schedule. Gantt charts are a highly visual representation of a project's tasks, with clear breakdowns of who's responsible for the work and when those tasks are due.
- **Work management tools:** There are many options available (e.g., Smartsheet, Asana, Jira, Trello). It is important to keep in mind that every company, project manager, and customer has a work management tool preference. These tools allow for collaboration and communication at a task level.

- **Kanban boards:** A visual tool used to manage tasks and workflows. They can be created on whiteboards, computer programs, etc. Tasks are written on cards and placed in columns representing progress.
- **Information included in tasks (or Kanban cards):**
 - **Task ID numbers or task names:** Makes it easy to find and reference a task.
 - **Task durations:** The amount of time you estimate that task should take.
 - **Start and finish dates:** Helps you track whether you are progressing on time.
 - **Who is responsible for what:** Role and responsibilities. Ideally, assign one owner per task/card.
- **Anatomy of a Kanban Card:**
 - **Front:** Title and unique identifier, Description of work (brief), Estimation of effort (e.g., small, medium, large), Who is assigned to the task.
 - **Back:** Start date, Blocked days (days your task may be halted), Finish date.

Module 3: Managing budgeting and procurement

Project Budgeting Basics

- **Project budget:** The estimated monetary resources needed to achieve the project's goals and objectives.
- **Milestone:** An important point within the project schedule that indicates progress and usually signifies the completion of a deliverable or phase of the project.
- **Forecast:** A cost estimate, or prediction over a period of time.
- Budget creation takes place in the **initiation phase** of your project.
- The budgeting process usually happens in conjunction with the scheduling process.
- **Reasons for a project (or value of budgeting):** Increase workforce productivity, Increase revenue, Attempt to save costs.
- A project manager must show the requested amount of money was used in order to secure enough budget for future projects.

Considerations When Creating a Budget

- When creating a budget, a project manager must account for:
 - Understanding stakeholder needs
 - Budgeting for surprise expenses
 - Maintaining adaptability
 - Reviewing and reforecasting throughout the project
- **Resource cost rates:** The cost of a resource.
- **Reserve analysis:** A method to check for remaining project resources.
- **Contingency budget:** Money that is included to cover potentially unforeseen events that aren't accounted for in a cost estimate.
- **Cost of quality:** Costs that are incurred to prevent issues with products, processes, or tasks. Includes:
 - Prevention costs
 - Appraisal costs
 - Internal failure costs
 - External failure costs

Direct vs. Indirect Costs

- **Direct costs:** These are costs for items that are necessary in order to complete your project. These costs can include:
 - Wages and salaries of employees and contractors
 - Materials costs
 - Equipment rental costs
 - Software licenses
 - Project-related travel and transportation costs
 - Staff training

- **Indirect costs:** These are costs for items which do not directly lead to the completion of your project but are still essential for the project team to do their work. They are also referred to as **overhead costs**. These costs can include:
 - Administrative costs
 - Utilities
 - Insurance
 - General office equipment
 - Security

Steps to Create and Manage a Budget

- **Budgeting steps:**
 - Break project into tasks
 - Estimate cost of each item
 - Add estimates together
 - Add contingency and tax
 - Seek approval from key stakeholders
- Monitoring the budget is crucial for a project manager to enforce accountability in terms of spending.

Managing Budget Changes and Cost Control

- **Cost control:** Practice where a project manager identifies factors that might impact their budget and then creates effective actions to minimize variances.
- Milestones are a great opportunity to re-review the budget, to identify if anything needs to be reset or revisited throughout the project.
- **Best practices for managing budget changes:**
 - Establish a sign off plan and inform the appropriate stakeholders of any changes that occur
 - Manage changes as they're made
 - Accept that budget misses will happen
 - Adequately account for, adapt, and manage your budget with that risk in mind

Understanding Expenses and Cash Flow

- **Cash flow:** The inflow and outflow of cash on your project. As a project manager, this is important to understand because you need funding (cash into your project) to keep your project running.
- **CAPEX (Capital Expenses):** An organization's major, long-term, upfront expenses, such as buildings, equipment, and vehicles. They are generally for assets that the company will own and keep. The company incurs these expenses because they believe they will create a benefit for the company in the future.
- **OPEX (Operating Expenses):** The short-term expenses that are required for the day-to-day tasks involved in running the company, such as wages, rent, and utilities. They are often recurring.

Procurement Basics

- **Procurement:** Obtaining all of the materials, services, and supplies required to complete the project.
- **Vendors:** Individuals or businesses who provide essential goods and services.

- **Contract Types:**

- **Fixed contracts:** Paid for when certain milestones are reached.
- **Time & materials contracts:** Paid monthly based on the hours worked and other fees associated with the work, like travel and meals.

Procurement Management Approaches

- **Agile procurement management:** Collaborative with both the project team and the end supplier. Emphasis is on the relationship between these parties. The project team plays a larger role in identifying what needs to be procured.
- **Traditional procurement management:** Focus on standard contracts with clear terms and deliverables. The project manager may be responsible for end-to-end procurement. Contracts may feature lengthy and extensive documentation.

Procurement Documents

- **Request for Proposal (RFP):** A document that outlines the details of the project.
- **Statement of Work (SoW):** A document that clearly lays out the products and services a vendor or contractor will provide for the organization.
- **Non-Disclosure Agreement (NDA):** A document that keeps confidential information within the organization.

The Procurement Process

1. **Initiating:** Planning what you need to meet your project goals
2. **Selecting:** Sourcing vendors, getting quotes, deciphering which vendors will fulfill your needs, and deciding which suppliers and vendors to use
3. **Contract writing:** Developing, reviewing, negotiating vendor contracts, and signing contracts
4. **Controlling:** Setting deadlines, evaluating performance, maintaining and ensuring quality, making payments, and ensuring vendors are paid
5. **Completing:** Measuring your success

Ethics in Procurement

- "Honesty, responsibility, respect, and fairness are the values that drive ethical conduct for the project management profession."
- Think critically when it comes to ethics in procurement. It is important to maintain ethics when partnering with vendors.
- **Potential unethical risks:**
 - Bribery or corruption
 - Sole-supplier sourcing
 - Interaction with state-owned entities
- **Test your ethics:** When facing an ethical dilemma, ask yourself questions in each category:
 - **Shame:** Would you be ashamed if someone knew what you did?
 - **Community:** Would you want your friends to know the decision you made?
 - **Legal:** Would you face legal action if you took this action?
 - **Situation:** Would your actions be justified in this situation?
 - **Consequence:** Would a negative outcome be worth your actions?

Module 4: Managing risks effectively

Risk and Issue Basics

- **Risk:** A potential event which can occur and can impact your project.
- **Issue:** A known or real problem that can affect the ability to complete a task.
- **Risk management:** The process of identifying and evaluating potential risks and issues that could impact a project.
- **Understanding provided by risk management:**
 - What could go wrong
 - Who you'll need to consult
 - How the risk could be mitigated
- Failing to plan for risk puts project goals, timelines, success criteria, and pivot plans at risk.
- Risks affect projects in unforeseen ways. When issues arise, keep calm, find the root cause, and solve it.

The 5-Step Risk Management Process

1. **Identify the risk:** Define potential project risks with the team.
2. **Analyze the risk:** Determine likelihood (probability) and potential impact to your project. Serious risks with a high probability pose the greatest threat.
3. **Evaluate the risk:** Prioritize risks based on the results of the risk analysis.
4. **Treat the risk:** Make a plan for how to manage and treat each risk (mitigation plans).
5. **Monitor and control the risk:** Assign team members to track, monitor, and mitigate risks.

Key Risk Assessment Concepts

- **Scope creep:** Changes, growth, and uncontrolled factors affecting project scope at any point after the project begins.
- **Risk register:** A table or chart containing a list of risks.
- **Risk assessment:** The stage of risk management where qualities of a risk are estimated or measured.
- **Probability and impact matrix:** A tool used to prioritize project risks.
- **Impact:** The damage a risk could cause if it occurs (high, medium, low).
- **Probability:** The likelihood that a risk will occur.
- **Inherent risk:** The measure of a risk calculated by its probability and impact.
- **Risk appetite:** The willingness of an organization to accept possible risk outcomes.
- **Single point of failure:** A catastrophic risk that has the potential to halt all work across a project.

Cause-and-Effect Diagrams

- **Cause-and-effect diagram (Fishbone/Ishikawa diagram):** Shaped like a fish skeleton, developed by Kaoru Ishikawa in the 1960s to measure quality control processes in the shipbuilding industry. It is a visual way to analyze the possible causes of an event or risk.
- **Fishbone Diagram Steps:**
 - Define the problem
 - Identify categories

- Brainstorm causes
- Analyze causes

Fishbone Ishikawa Diagram Concept

Risk Types and Task Dependencies

- **Risk Types:**
 - **Time risk:** Possibility that tasks take longer than anticipated.
 - **Budget risk:** Possibility that costs increase due to poor planning or expanding scope.
 - **Scope risk:** Possibility that a project won't produce the outlined results.
 - **External risk:** Risks from factors outside company control (e.g., regulatory changes or tax increases).
- **Dependency:** A relationship between two project tasks, where the start or completion of one depends on the start or completion of the other. Dependencies connect tasks and are a huge source of risk to schedule, budget, or outcomes.
- **Internal dependencies:** Dependencies within the project that you and your team have control over.
- **External dependencies:** Dependencies that you have no control over.
- **Four Common Types of Dependencies:**
 - **Finish to Start (FS):** Task A must be completed before Task B can start. (Most common).
 - **Finish to Finish (FF):** Task A must finish before Task B can finish.
 - **Start to Start (SS):** Task B can't begin until Task A begins (run in parallel).
 - **Start to Finish (SF):** Task A must begin before Task B can be completed.
- **Dependency graph:** A visualization of dependencies and the workflow to identify associated risks.

AI Tools and Risk Mitigation Strategies

- **Using AI to discover risks:** Ask what can go wrong, generate/reformat content, use your own judgment to verify, and brainstorm mitigation plans.
- **Four Core Risk Mitigation Strategies:**
 - **Avoid:** Sidestep the situation as a whole (e.g., using a widely available alternate seed).
 - **Minimize:** Try to minimize the catastrophic effects (e.g., using both the original and a neighboring supplier).
 - **Transfer:** Shift the responsibility of handling the risk to someone else (e.g., purchasing from a local supplier who absorbs regulatory risks).
 - **Accept:** Accept risk as normal cost of doing business.
 - **Active acceptance:** Set aside extra funds or contingency reserves.
 - **Passive acceptance:** "Do nothing" (reasonable for smaller risks, but not recommended for single points of failure).
- **Decision tree:** A flowchart that helps visualize the wider impact of a decision on the rest of a project.
- **Risk management plan:** A living document containing information regarding high-level risks and mitigation plans. It should be updated regularly to add newly-identified risks, remove irrelevant ones, and include mitigation changes.

- **Stakeholder Risk Communication:**

- Failing to communicate risks means stakeholders are less equipped to help and trust is eroded.
- Communicate early and often about medium- and high-level risks.
- High-level risks require thorough and direct communication; the specific method depends on the severity.

Module 5: Organizing communication and documentation

Communication Basics

- **Communication:** The flow of information, which includes what is shared, how it's shared, and with whom.
- **Effective communication characteristics:** Clear, Honest, Relevant, Frequent.
- **Types of communication:** Meetings, Emails, Phone calls, Written documents, Formal presentations.
- Communication needs to happen throughout the entire life cycle of the project.
- **Communication plan:** Organizes and documents the process, types, and expectations of communication for the project.
- **What a communication plan details:**
 - What needs to be communicated
 - Who needs to communicate
 - When communication needs to happen
 - Why and how to communicate
 - Where the information being communicated is stored
- **Planning communications upfront:**
 - Improves communication overall
 - Keeps people engaged and motivated
 - Gets stakeholders involved in effective conversations
- **Modes of information processing:** Visuals, Listening, Reading and analysis, Talking with others.
- **Evaluation checklist for communications:** What is working? What is not working? Where can improvements be made?

Creating a Communication Plan

- **Key Steps (Identify, Identify, Identify):**
 - **Project stakeholders:** Have you created a RACI chart or stakeholder map? Who is the audience? Who needs to be informed at different points?
 - **Communication frequency and method:** When/how often to check in? Preferred methods? How much detail is needed?
 - **Goals:** What is the communication goal? Is a response needed? Encouraging engagement or simple updates?
 - **Barriers:** Identify time zones, language barriers, response time limits (especially for executives), privacy, or internet access issues.

Documentation Organization and Storage

- Centralized and clearly labeled plans make communication quicker, easier, and more streamlined, providing visibility, accountability, and continuity.
- **Continuity:** Up-to-date plans ensure there's no room for miscommunication, so that even in the project manager's absence, the project can carry on.

- **Access control:** Not everyone needs full access to project documentation. Summarize relevant information for those who need to stay informed of final outcomes, but don't need all the background details.
- **Knowledge management:** A way of ensuring that project data can be accessed in the future by others who need it for informing decisions or planning similar projects.
- **Pro tips:**
 - Determine what kind of information to share with whom and when.
 - Protect sensitive data from unauthorized viewers.
- **Need-to-know basis:** Telling someone facts they need to know at the time they need to know them, and nothing more.
- **Personally Identifiable Information (PII):** Information that could be used on its own to directly identify, contact, or precisely locate an individual.
- **Personally Identifiable Information (PII) Examples:**
 - Email addresses
 - Mailing addresses
 - Phone numbers
 - Precise locations
 - Full names or usernames
- Have all project resources documented and linked so that anyone can access what they need quickly.